

Week 06: Effects and do notation

Functional Programming in Lean

Learning goals

- ▶ Read `Option`, `Except` and `IO` result types
- ▶ Sequence computations with `do`
- ▶ Keep errors explicit in a function signature

Central idea

An effectful computation returns a value together with a declared possibility such as failure.

Lean example

```
def safeDivide (x y : Nat) :  
  Except String Nat :=  
  if y == 0 then .error "division by zero"  
  else .ok (x / y)
```

What to notice

- ▶ Except `String Nat` contains a result or an error
- ▶ `do` propagates an error without repeated matches
- ▶ `IO` marks interaction with the external world

Live coding

1. Complete `safeDivide`
2. Write `ratioOfSums` with `do`
3. Compare success and failure with `#eval`

Pause and predict

What does `ratioOfSums 10 2 0 0` return?

Exercise connection

Open `exercises/Week06.md` after the lecture.

The complete example is `Course/Week06.lean`; the fill-in version is `Templates/Week06.lean`.